

Chapter 1

Proximity Service

In this chapter we design a proximity service.

- A proximity service is used to discover nearby places such as restaurants, hotels, theaters, museums, etc., and is a core component that powers features like finding the best restaurants nearby on Yelp or finding k-nearest gas stations on Google Maps.
- Figure 1.1. shows the user interface via which you can search for nearby restaurants on Yelp [1]
- Note the map tiles used in this book are from Stamen Design [2] and data are from OpenStreetMap [3]

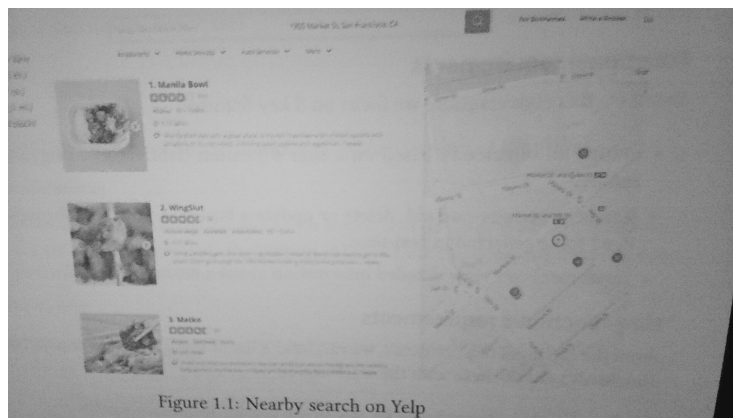


Figure 1.1: Nearby search on Yelp

Nearby search on yelp

Step 1 - Understand the Problem and Establish Design Scope

Yelp supports many features and it is not feasible to design all of them in an interview session, so it's important to narrow down the scope by asking questions

The interactions between the candidate and interviewer could look like this:

- Candidate: Can a user specify the search radius? If there are not enough businesses within the search radius, does the system expand the search?
- Interviewer: That's a great question. Let's assume we only care about businesses within a specified radius. If time allows, we can then discuss how to expand the search if there are not enough businesses within the radius
- Candidate: Can a user change the search radius on the UI?
- Interviewer: Yes, we have the following options: 0.5km (0.31 mile), 1km (0.62 mile), 2km (1.24 mile), 5km (3.1 mile), and 20 km (12.42 mile)
- Candidate: How does business information get added, deleted, or updated? Do we need to reflect these operations in real-time?
- Interviewer: Business owners can add, delete or update a business. Assume we have a business agreement upfront that newly added/updated businesses will be effective the next day.
- Candidate: A user might be moving while using the app/website, so the search results could be slightly different after a while. Do we need to refresh the page to keep the results up to date?
- Interviewer: Let's assume a user's moving speed is slow and we don't need to constantly refresh the page

Functional requirements

Based on this conversation, we focus on 3 key features:

- Return all businesses based on a user's location (latitude and longitude pair) and radius
- Business owners can add, delete, or update a business but this information doesn't need to be reflected in real time
- Customers can view detailed information about a business